

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	G. Raju	Reg. No.:	192411065
Section / Batch:	2024 / CSE	Date:	

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I, G. Raju (192417065) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Raju

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Name :- G. Raju

COURSE CODE :- CSA0618

reg No :- 192411065

COURSE NAME :- DAA

CO2 - AT1

1. bubble sort :-

Problem :- sort the array [45, 12, 78, 34, 23] in Ascending order using bubble sort.

Method :- bubble sort compares adjacent element & swaps them if they are in the wrong order.

Steps :- initial Array

45 72 78 34 23

Pass 1 :- 45 & 12 → swap

12 45 78 34 23

45 & 78 → No swap

78 & 34 → swap

12 45 34 78 23

78 & 23 → swap

12 45 34 23 78

Pass 2 :- 45 & 34 → swap

12 34 45 23 78

45 & 23 $\Rightarrow$ swap

12 34 23 45 78

Pass 3:- 34 & 23 $\Rightarrow$ swap

12 23 34 45 78

Pass A:- Already sorted

12 23 34 45 78

Result:- sorted array

[12, 23, 34, 45, 78]

Comparison = 70

Swap = 6

Time complexity = $O(n^2)$

2) Linear Search

Problem:- find 34 is

23 45 12 67 34 89 10

method:- compare each element one by one

Steps:

Comparison

Result:

23 = 34?

No

45 = 34?

No

12 = 34?

No

67 = 34?

No

34 = 34?

Yes

element found

Position = 3

index = 4

Comparison = 5

Time Complexity = $O(n)$

③ Brute force string matching

Problem:- Text = ABAB ABCA BABABC AB

Pattern = ABABC

Method:- Compare the Pattern at every possible position.

Steps:-

Position

Result:-

0

mis match

1

mis match

2

match

3

mis match

4

mismatch

5

mismatch

6

mismatch

7

mismatch

8

match

occurrences = 2 (position 2 & 3)

Comparison = 23

Time Complexity = $O(n \times m)$

4) Maximum Subarray

Problem :- Array

$[-2, 3, 7, -7, -2, 7, 5, 0]$

Method :

Calculate the sum of every possible subarray

& choose the maximum

Import

$$[-2] = -2$$

$$[3] = 3$$

$$[3, 7] = 10$$

$$[3, 7, -7] = 3$$

$$[3, 7, -7, -2] = 1$$

$$[3, 7, -7, -2, 7, 5] = 20 \leftarrow \text{Maximum}$$

$$[4, 1, -2, 1, 6] = 7$$

$$[1, 5] = 6$$

Maximum sub array

$$[3, 4, -2, -2, 1, 5]$$

$$\text{Maximum sum} = 70$$

$$\text{total sub array} = 36$$

$$\text{time complexity} = O(n^2)$$

5. Nearest Neighbour distance

Problem points :-

$$A(7, 2) \quad B(4, 6) \quad C(7, 8) \quad D(2, 3) \quad E(5, 5)$$

Method :

find the distance b/w every pair using

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Calculation

<u>Pair</u>	<u>distance</u>
A-B	5.00
A-C	8.49
A-D	7.41
A-E	5.00
B-C	3.67

B-E	7.47
C-D	7.07
C-E	3.61
D-E	3.61

Result:

→ Nearest Pairs

① A-D

② B-E

→ Minimum distance = 7.47

→ Comparison = 10

→ Time Complexity = $O(n^2)$